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Key Points:

- We developed a novel dynamic index to quantify hydro-sediment connectivity in permafrost rivers
- This study reveals increasing trends (2000–2022) and intra-annual shifts in sediment connectivity
- Cryo-Functional Sediment Connectivity links hillslope to channel dynamics, distinguishing connected versus disconnected sediment sources in cold regions

Supporting Information:

Supporting Information may be found in the online version of this article.

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Cryo-FSC: A Dynamic Functional Connectivity Index Reveals Spatio-Temporal Shifts in Sediment Transport in the Permafrost-Dominated River Systems

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Abstract Cryospheric degradation in cold regions has been fundamentally altering hydrological and geomorphic processes, intensifying sediment dynamics with profound implications for water resources management. Existing sediment connectivity indices lack the dynamic resolution to capture the intricate, seasonally evolving processes unique to cold regions. Here, we introduce the Cryo-Functional Sediment Connectivity (Cryo-FSC) index, integrating key cryospheric (e.g., snowmelt, freeze-thaw state) and surface processes (e.g., normalized river channel steepness) to dynamically quantify hillslope-channel coupling in the Source Region of the Yangtze River (SRYP). Our analysis reveals that Cryo-FSC exhibits significant spatio-temporal heterogeneity, with an overall increasing trend over 2000–2022, particularly in the upstream glacierized sub-basins (e.g., Tuotuo River: $12.64 \pm 3.83\%/yr$, $P < 0.001$). Crucially, Cryo-FSC successfully captures a distinct intra-annual seasonality, displaying peak connectivity during the thaw season (June–September) and consistently preceding observed suspended sediment flux peaks by 3–5 days. We further demonstrate Cryo-FSC's utility as a spatial framework for interpreting localized erosion features such as retrogressive thaw slumps, helping distinguish potentially connected source areas from disconnected deposits. This dynamic index offers an unprecedented tool for deciphering the complex sediment cascades in rapidly changing cryospheric landscapes, providing crucial insights for regional sediment management and landscape evolution.

Plain Language Summary In cold mountainous regions like the Tibetan Plateau, global warming is rapidly melting glaciers and thawing frozen ground. This thawing destabilizes the soil, making it easier for rain and meltwater to transport large amounts of sediment into major rivers. To better understand it, we developed a new tracking tool called the Cryospheric Functional Sediment Connectivity (Cryo-FSC) index. Unlike older tools that only looked at the static shape of the land, our new index acts as a dynamic map. It explicitly accounts for seasonal changes, such as the timing of snowmelt and the summer thawing of the frozen ground. By applying this tool to the Source Region of the Yangtze River over a 22-year period (2000–2022), we discovered that the pathways carrying sediment into the rivers have become increasingly active, especially in areas with shrinking glaciers. Crucially, our tool successfully captures the seasonal timing of erosion, showing peak activity about 3–5 days before the largest surges of sediment actually appear in the river. This new approach provides a valuable way to manage water resources and predict landscape changes in rapidly warming cold regions worldwide.

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1. Introduction

Accelerated climate change is profoundly altering sediment transport from landscapes to river networks, impacting global water quality, reservoir longevity, and downstream ecosystems (Bracken et al., 2013, 2015; Heckmann et al., 2018). Quantifying the efficiency of these water and sediment cascades—called sediment connectivity—is therefore essential for predicting landscape evolution and mitigating associated risks (Michalek et al., 2023; Partington et al., 2017). Beyond merely mapping static topographical pathways, an advanced connectivity model can answer broader questions concerning hydrogeomorphic shifts by revealing exactly what portions of the landscape have the highest potential to transport sediment fluxes to rivers, and how these potentials vary in space and time. However, the quantitative, spatiotemporal characterization of dynamic connectivity at the basin scale and beyond remains exceedingly rare (e.g., López-Vicente and Ben-Salem, 2019; Wu et al., 2023). This limitation primarily stems from prohibitive computational demands, the scarcity of high-resolution, publicly available data products, and profound regional heterogeneity (East & Sankey, 2020). This challenge is particularly acute within Earth's high-altitude cryosphere, where amplified warming drives dramatic yet poorly understood shifts in sediment dynamics and connectivity (Li et al., 2021).

Prevailing approaches often rely on structural connectivity (IC) based on static topography to delineate potential sediment pathways (Fryirs, 2013; Harvey, 2001; Messenzehl et al., 2014; Najafi et al., 2021; Ortíz-Rodríguez et al., 2019). However, these static “snapshots” fundamentally fail to capture the crucial process dynamics governing sediment mobilization and transport, especially under non-stationary conditions, severely limiting their predictive power (Najafi et al., 2021). While the development of functional sediment connectivity (FSC), incorporating factors like rainfall and soil properties (López-Vicente and Ben-Salem, 2019), marked a conceptual step forward toward capturing process dynamics, these indices were primarily developed for temperate environments and are ill-suited for the cryosphere (Wu et al., 2023). They largely neglect the unique, dominant, and highly non-linear processes governing cold region sediment dynamics.

These neglected processes include: (a) intense, pulsed sediment release driven by glacier and snow meltwater (Beylich et al., 2017); (b) the profound influence of freeze-thaw cycles on erodibility, infiltration, and slope stability (J. Li et al., 2023); and (c) highly seasonal vegetation cover changes that drastically alter surface roughness and resistance (D. Li et al., 2024). These factors drive exceptionally rapid environmental responses; for instance, the Tibetan Plateau (TP), Earth's “Third Pole”, is warming at over twice the global average rate (J. Li et al., 2024; Yao et al., 2012), contributing to dramatic increases in observed sediment flux in major rivers like the Yangtze (e.g., >10.76% increase documented at the Tuotuo River headwaters (Li et al., 2020)). The absence of a framework integrating these cryo-specific drivers means our ability to quantitatively understand and predict sediment connectivity, and thus downstream sediment loads and associated hazards, in these globally significant, rapidly transforming regions remains critically hampered (Wang et al., 2011; Wasson et al., 2022).

Here, we address this critical knowledge gap by developing and applying a novel Cryospheric Functional Sediment Connectivity (Cryo-FSC) index. Uniquely, Cryo-FSC explicitly integrates key cold-region dynamics neglected in previous formulations: (a) temporally varying glacier or snow meltwater contributions derived from enhanced, regionally calibrated temperature-index models or energy balance approaches; (b) rapid intra-annual vegetation shifts quantified using high-resolution satellite-derived seasonality metrics; and (c) the transient effects of active layer freeze-thaw status on soil erodibility and runoff generation, parameterized via surface temperature proxies or simplified thermal models (Bayard et al., 2005; Lamoureux et al., 2014; Rodzik et al., 2009). We implement and validate Cryo-FSC across the source region of the Yangtze River (SRYR) on the Tibetan Plateau, a globally significant cryospheric basin. Ultimately, the central takeaway from our modeling framework and results is a significantly improved, dynamic understanding of exactly which portions of the landscape have the highest potential to transport sediment fluxes to fluvial systems, and precisely how these critical transport potentials vary in space and time. By isolating these highly active sediment routing pathways, the Cryo-FSC index acts as a powerful spatial routing engine that, when integrated with future quantitative erosion models, will enable robust predictions of actual sediment yields and landscape sensitivity in rapidly evolving cold regions worldwide.

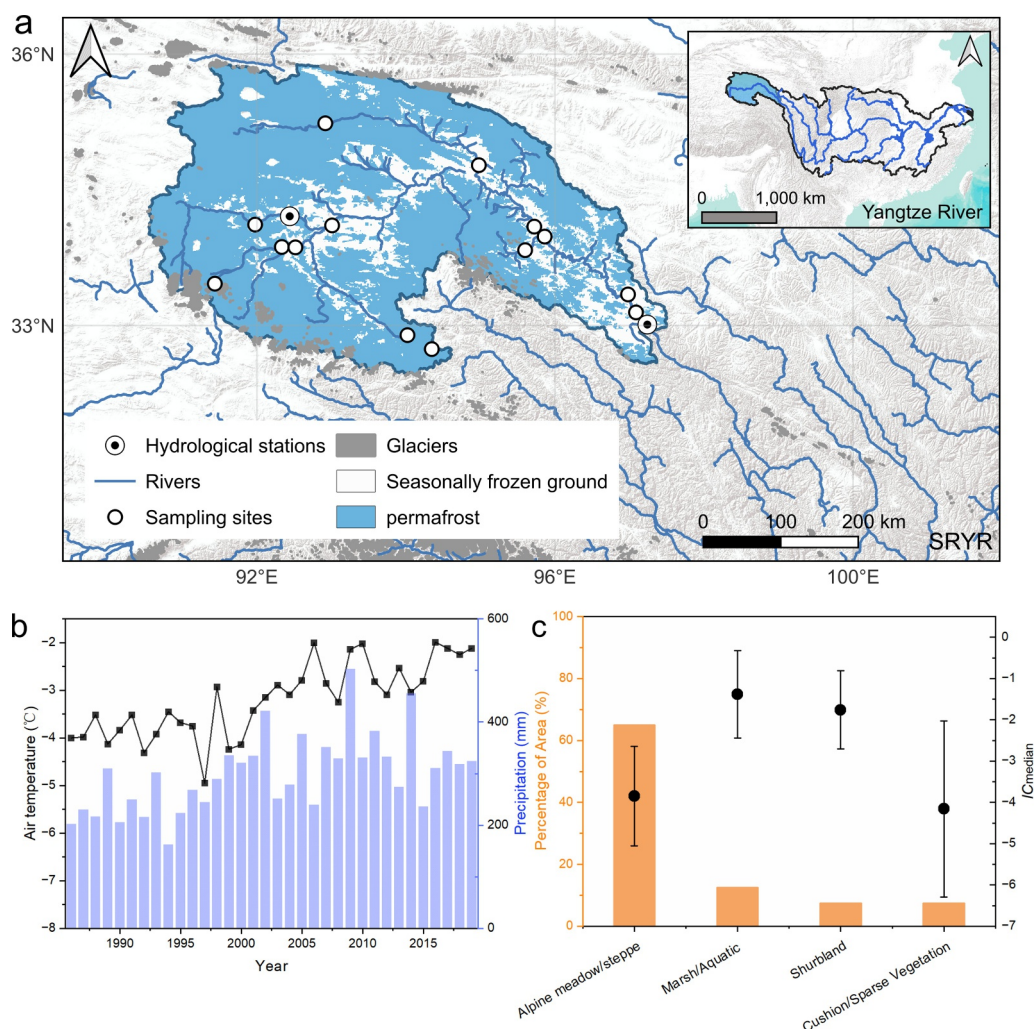


Figure 1. Basic information includes geomorphology, river systems, climate, land cover, and structural connectivity in the Source Region of the Yangtze River (SRYR). (a) Location, rivers, permafrost, glacier distribution, and sampling sites. The inset panel in the upper right corner illustrates the location of the SRYR. (b) Increase in the air temperature and precipitation. Data were obtained from daily field observations at the Tuotuo River Hydrological Station, Qinghai Provincial Bureau of Hydrology and Water Resources. (c) Coverage areas of the main land cover types in the SRYR and corresponding changes in structural connectivity (IC). Land use data were sourced from Xiao Zhang et al. (2021), while structural connectivity data were derived from the structural connectivity data set over High Mountain Asia developed by Li et al. (2025).

2. Materials and Methods

2.1. Study Area

The SRYR, a $13.8 \times 10^4 \text{ km}^2$ basin situated between the Kunlun and Tanggula Mountains ($32^\circ 30' - 35^\circ 35' \text{N}$, $90^\circ 43' - 96^\circ 45' \text{E}$) on the TP, exhibits a high-west, low-east topography with diverse landforms (Figure 1). Crucially, it represents a typical high-altitude permafrost basin, with permafrost underlying 77.3% of its area (Song et al., 2020) and undergoing significant degradation (e.g., thickening active layers) driven by accelerated regional warming (Dai & Liu, 2013). Geologically, the area comprises sedimentary, silicate, and carbonate rocks. The climate is subarctic semi-arid to semi-humid, characterized by cold, dry conditions (mean annual temperature: $-6^\circ \text{C} - 4^\circ \text{C}$) and summer-concentrated precipitation (Liu et al., 2020; Zhang et al., 2023).

Dominated by natural ecosystems with relatively limited, albeit increasing, anthropogenic influence, the SRYR serves as an ideal natural laboratory for investigating hydrological connectivity dynamics (J. Li et al., 2023; Ma et al., 2014). The landscape primarily features alpine meadows (covering $\sim 60\%$, dominated by *Kobresia* spp.) and alpine steppes, overlying generally thin alpine soils with distinct zonation. The region possesses a dense river

network (>80 rivers, including the major Tuotuo and Tongtian Rivers), extensive lakes (~800 km²), and significant glacier cover (~1,500 km²; Juez et al., 2021). Governed by precipitation and pervasive freeze-thaw cycles, the annual hydrological regime exhibits a characteristic double-peaked hydrograph, with peak discharge in summer months and minimal, baseflow-dominated flow during the extended cold season (December–March). More details can be found in Figures S1–S3 and Table S1 in Supporting Information S1.

2.2. The Cryospheric Functional Sediment Connectivity (Cryo-FSC) Index

To quantitatively assess spatio-temporal sediment connectivity dynamics within the cryosphere-dominated SRYR, we developed the Cryo-FSC index. This index builds upon the aggregated index of sediment connectivity (AIC) framework proposed by López-Vicente and Ben-Salem (2019), which itself adapted the original IC concept by Borselli et al. (2008). The AIC framework allows incorporating multiple spatially and temporally variable factors influencing connectivity. However, standard AIC formulations lack explicit representation of key cryospheric processes critical: (a) pulsed water inputs from snow and glacier melt, (b) rapid intra-annual vegetation dynamics characteristic of alpine environments, and (c) the profound impact of soil freeze-thaw cycles on surface erodibility and permeability.

The Cryo-FSC index addresses these limitations by retaining the core logarithmic structure of IC/AIC while incorporating cryosphere-specific dynamic factors into an aggregated weighting component (Cryo-AWC). The index for each grid cell $Cryo_k$ is calculated as:

$$Cryo_k = \log_{10} \left(\frac{D_{up,k}}{D_{dn,k}} \right) \quad (1)$$

where $D_{up,k}$ is the upslope component representing the potential for downward routing of sediment and water, and $D_{dn,k}$ is the downslope component representing the impedance encountered along the flow path to the defined target. Higher values of the Cryo-FSC index indicate greater potential sediment transport efficiency, implying stronger functional connectivity between upslope sources and downslope areas. The subsequent sections detail the calculation of the constituent factors contributing to $D_{up,k}$ and $D_{dn,k}$.

2.2.1. Upslope Component Factors

In our implementation, the upslope component $D_{up,k}$ is determined by the potential of downward movement of the upslope flow and sediment, depending on local conditions, slope and size of the drainage area. Specifically, we employed a formulation using the total basin area:

$$D_{up,k} = \overline{W}_i \cdot \overline{RT} \cdot \overline{C}_i \cdot \overline{FT}_i \cdot \overline{K}_p \cdot \overline{S} \cdot \sqrt{A_k} \quad (2)$$

Here, A_k is the upslope contributing area (m²), $\overline{S}$ is the average slope gradient (m/m) of A_k , and $\overline{S}$ is the local slope gradient of cell i . The factors $\overline{W}_i$, $\overline{RT}$, $\overline{C}_i$, $\overline{FT}_i$, and $\overline{K}_p$ are normalized indices (ranging from 0 to 1) representing water erosivity, residual topography, vegetation cover and management, freeze-thaw state, and intrinsic soil permeability, respectively. The bar notation (e.g., $\overline{W}_i$) indicates the average value of the factor over the contributing area A_k . Notably, in our Cryo-FSC formulation, we utilized the total catchment contributing area (A_k) rather than the pixel-specific upslope area commonly found in IC variants. A_k was computed by summing the upslope area (km²) from the MERIT-Hydro data set over the entire Area of Interest (AOI) and converting the result to square meters. Calculation details for each factor are provided below.

The surface weighting factor synthesizes the impact of multiple factors influencing surface erosion and runoff generation. Its calculation draws inspiration from the Revised Universal Soil Loss Equation (RUSLE) and is derived as the product of several sub-factors. For clarity, dynamic factors are designated with a subscript “ i ” to indicate their monthly temporal evolution, while static factors remain unsubscripted.

Water Erosivity Factor ($\overline{W}_i$): This factor integrates erosive potential from both rainfall and meltwater (including glacier and snow meltwater) in cold regions, including Rainfall Component, Meltwater Component, and Combination and Normalization Component. It is important to note that the Cryo-FSC index operates within a coupled modeling framework. While spatial accumulation and sediment routing are executed on the Google Earth Engine

(GEE) platform, the underlying cryospheric forcing variables are process-based products simulated offline. We utilized the runoff module of the modular, distributed hydro-thermal coupled (DHTC) hydrological model for cold regions, originally developed by Guo et al. (2024), to simulate the meltwater runoff and generate a gridded spatial product. To address the spatial resolution mismatch between the DHTC simulation outputs and the high-resolution topographic baseline in GEE, continuous hydrological fluxes, such as effective melt runoff (M_{snow}), were downscaled using bilinear interpolation. This approach mitigates blocky boundary artifacts and ensures the hydro-geomorphic continuity necessary for sediment routing (Figure S7 in Supporting Information S1). For the rainfall component, daily precipitation data are sourced from meteorological stations and supplemented by the simulated surface runoff data (L. Wang et al., 2021). The monthly rainfall factor was calculated following Islam (2022):

$$R = 1.73 \times 10^{(1.5 \times \log(P_m^2/P_a) - 0.08188)} \quad (3)$$

where R is the erosivity by rainfall in megajoules (MJ) in $\text{mm/ha} \cdot \text{h}^{-1} \cdot \text{y}^{-1}$, P_m is the precipitation in the month and P_a is the precipitation in a year.

Meltwater from snow was simulated using the temperature-index (degree-day) module embedded within the DHTC model. This module estimates melt rates by integrating both air temperature and incident shortwave radiation, following the generalized equation:

$$M_{\text{snow}} = \text{DDF}_{\text{snow}} \cdot (T_{\text{air}} - T_{\text{snow}}) + \text{SRF}_{\text{snow}} \cdot \text{NS}_{\text{snow}} \quad (4)$$

where DDF_{snow} is the degree-day factor ($\text{mm} \cdot ^\circ\text{C}^{-1} \cdot \text{day}^{-1}$), T_{air} is the air temperature, T_{snow} is the threshold of snow temperature, and NS_{snow} is the net shortwave radiation ($\text{W} \cdot \text{m}^{-2}$). This formulation enables realistic meltwater generation in cold regions, particularly in data-sparse permafrost basins, by capturing key radiative and thermal controls on snowmelt. Glacier meltwater was also simulated using the same degree-day method as snowmelt, but restricted to glacierized areas through the application of a glacier mask.

Furthermore, the meltwater generated from each grid cell was then converted into a unitless measure of meltwater-induced erosivity potential. To achieve this, we assumed that meltwater possesses a baseline erosive power proportional to both its volume and flow intensity. Specifically, the erosivity E_{melt} was estimated as:

$$E_{\text{melt}} = k \cdot \frac{Q_{\text{melt}}}{A} \quad (5)$$

where Q_{melt} is the simulated meltwater runoff (mm/month), A is the grid area (km^2), and k is an empirical scaling coefficient derived from observed melt-runoff erosivity relationships in cold-region catchments. This approach is conceptually analogous to rainfall kinetic energy-based indices (e.g., EI30), but adapted for thermal-driven hydrological processes (López-Vicente and Ben-Salem, 2019).

It is important to note that while Equation 5 applies a unified parameterization to convert total meltwater volume into local erosivity, the framework inherently distinguishes the geomorphic impacts of snowmelt and glacier melt through its spatial routing topology. Snowmelt is spatially distributed across hillslopes, where an increase in E_{melt} effectively reduces routing impedance and activates new hillslope-channel sediment pathways. In contrast, glacier melt is typically concentrated at glacier termini or proglacial zones. Because these locations are already topologically adjacent to or part of the established channel network, the local increase in E_{melt} primarily acts to maintain these pathways as highly efficient transport corridors, reflecting their naturally channelized and highly connected state. Thus, the contrast between snowmelt and glacier melt in Cryo-FSC arises from their spatial distribution and routing context rather than from separate erosivity formulations.

Finally, the monthly total water erosivity was subsequently calculated as the sum of rainfall- and meltwater-induced erosivity components ($W_t = R + E_{\text{melt}}$). To account for long-term spatial and seasonal variability, all values across the domain and period were normalized into the range (0.001, 1), thereby generating a gridded water erosivity index time series suitable for spatial analysis and erosion modeling.

Residual Topography (RT): RT was parameterized based on the standard deviation of slope gradients calculated within a 5×5 moving window (corresponding to a 30 m spatial resolution), serving as a proxy for surface roughness. The resulting values were normalized and then inverted using the equation:

$$RT = 1 - SD_{\text{slope_norm}} \quad (6)$$

where $SD_{\text{slope_norm}}$ refers to the normalized standard deviation of local slope gradients and serves as an indicator of relative surface roughness. Higher values indicate more rugged terrain, while lower values correspond to smoother and more level surfaces.

Dynamic Vegetation Factor (C_t): This factor reflects the influence of land cover for this region. The C-factor was derived using the relationship between land cover percentage (estimated from NDVI/EVI) and soil loss ratio (SLR) as described by (Wu et al., 2023):

$$SLR_i = e^{-0.0418 \times (TC_i - 5)} \quad (7)$$

$$C_i = SLR_i \cdot \frac{EI_i}{EI_a} \quad (8)$$

where SLR_i is the soil loss ratio in month i , TC_i is the total percentage of ground cover, C_i is the cover management factor for month i , and EI_i and EI_a are the monthly and annual rainfall erosivity, respectively.

Freeze-Thaw Factor (FT_t): This factor explicitly modulates connectivity based on the thermal state of the active layer. It was derived using daily mean land surface temperature (LST) from DHTC Heat transport module (Guo et al., 2024), which resolves coupled energy and moisture dynamics across multiple soil layers under freeze–thaw conditions. Satellite-based surface temperature and the global freeze–thaw index were used only for auxiliary calibration and consistency checks, rather than as direct inputs to the FT_t classification (Peng et al., 2019). These variables characterize the physical state of the hydro-thermal regime (e.g., subsurface thermal conduction and ice–water phase changes) that cannot be directly captured by satellite sensors alone. A threshold-based classification scheme was applied to each grid cell based on the surface layer's thermal regime.

Specifically, cells were classified as “frozen” (e.g., land surface temperature (LST) < 0°C for 3 consecutive days), “thawed” (e.g., LST > 0°C for 3 consecutive days), or “transitioning.” Unlike continuous meltwater fluxes, the FT_t relies on categorical states. Therefore, when integrating these simulated thermal states into the high-resolution GEE routing framework, the layers were reprojected using nearest-neighbor assignment. This resampling protocol preserves the discrete physical thresholds and state boundaries of the frozen and thawed ground.

Each thermal state was then assigned a corresponding FT_t value to represent its relative control on surface and subsurface flow continuity, with frozen soils assigned low values (e.g., 0.001) due to limited permeability, and thawed soils assigned high values (up to 1) to reflect enhanced infiltration and lateral connectivity. Transitioning states were assigned intermediate values based on empirical relationships as follows:

$$FT_t(LST) = \begin{cases} 0.001 & \text{LST} \leq 0^\circ\text{C}, \geq 3 \text{ consecutive days, Frozen} \\ \frac{1}{1 + \exp(-k(LST \pm 1))} & \leq 3 \text{ consecutive days, Transitioning} \\ 1 & \text{LST} \geq 0^\circ\text{C}, \geq 3 \text{ consecutive days, Thawed} \end{cases} \quad (9)$$

where LST represents the daily or monthly mean land surface temperature (°C). K is an empirical parameter that controls the steepness of the transition phases in the interval. The resulting FT_t index ranges from 0.001 to 1, and serves as a dynamic scaling factor within the broader connectivity framework, capturing seasonal and spatial variations in thermal controls on hydrological linkages.

The intrinsic soil permeability factor (K_p): It represents baseline impedance to water movement associated with inherent soil physical properties, independent of freeze–thaw conditions. It was derived from soil available water content data, which were obtained from HiHydroSoil v1.2 data set (De Boer, 2016). Following López-Vicente and Ben-Salem, (2019) and Wu et al. (2023), we used available water content as a proxy for intrinsic permeability.

To align with the conceptualization of impedance, the water content values were also normalized and inverted so that higher K_p values indicate lower intrinsic permeability.

Slope length and steepness (S) and Contributing Area (A): Calculated using the D-infinity algorithm (Tarboton, 1997). And slope values were normalized to the range from 0.001 to 1.

2.2.2. Downslope Component Factors

The downslope component $D_{dn,k}$ quantifies the probability of sediment reaching a downslope sink from upslope cell k along the flow path, accounting for cumulative flow length, terrain resistance, and local hydrological and geomorphic conditions:

$$D_{dn,k} = \sum_{i=k}^{n_k} \frac{d_i}{AWC_i} \quad (10)$$

$$AWC_i = Wt_i \cdot RT_i \cdot CT_i \cdot FT_i \cdot Kp_i \cdot S_i \quad (11)$$

Where d_i represents the length of the flow path across the i -th pixel along the steepest descent direction. Depending on the flow routing direction, it equals either the cell resolution (for orthogonal flow) or the cell resolution multiplied by $\sqrt{2}$ for diagonal flow. Wt_i is the water erosivity factor. RT_i is residual topography. CT_i is the land cover factor. FT_i is the freeze–thaw factor. Kp_i is intrinsic soil permeability, and S_i is slope. Each of these indices was calculated as described in the preceding sections.

2.2.3. Implementation and Target Definition

In this study, we focused on calculating dynamic connectivity between hillslopes and the river network in the SRYR, using both local drainage points and basin outlets as target nodes. We first delineated the sources of the mainstream and tributaries within the study area using the GEE platform. These locations were then used to define flow accumulation outlets for sub-basin segmentation and the identification of upslope and downslope contributing units. Subsequently, we applied the Deterministic Infinity flow algorithm to determine flow direction, using the average value of the flow accumulation raster as a threshold for defining upslope drainage areas. Based on the derived flow direction and accumulation grids, contributing areas were calculated iteratively for each grid cell.

The Cryo-FSC maps were generated by integrating the above components into a modified version of the Borselli sediment connectivity index (Ortiz-Rodríguez et al., 2019), incorporating cryosphere-specific weighting factors as defined in this study (Figure 2). Monthly maps of median, mean, and total Cryo-FSC values were produced for the period 2000–2022. All input and output layers were resampled or computed at a consistent spatial resolution of 30 m.

2.3. Data Collection and Pre-Processing

The calculation of the Cryo-FSC index relied on a comprehensive suite of data sets, including air temperature (T_{air} , °C), ground surface temperature (T_{sfc} , °C), precipitation (P , mm), surface runoff (d , mm), snow and ice melt runoff, subsurface soil temperature profiles, land cover distribution, and soil physical properties. For basin-scale assessment, air temperature (T_{air}) was obtained from the China Meteorological Forcing Data set (CMFD; hourly, $0.1^\circ \times 0.1^\circ$ resolution; <https://doi.org/10.11888/AtmosphericPhysics.tpe.249369.file>), while ground surface temperature (T_{sfc}) was extracted from ERA5-Land reanalysis (hourly, $0.1^\circ \times 0.1^\circ$ resolution; <https://cds.climate.copernicus.eu>).

Precipitation input was constructed by merging three data sources using the Triple Collocation Method: ERA5-Land, CMFD, and gridded precipitation interpolated from China Meteorological Administration (CMA) gauge stations via thin plate spline functions (<https://data.cma.cn>). The merged product served as the precipitation input for the hydrological modeling. Gridded data sets for surface runoff, snow/ice melt runoff, and soil parameters (including particle size distribution, bulk density, and organic matter content for each soil layer) were obtained from outputs of the DHTC model. The accuracy of these process-based simulated fields was rigorously validated against multi-source satellite observations, and the data sets were temporally gap-filled to ensure continuous coverage over the 2000–2022 study period (Figures S7–S9 in Supporting Information S1).

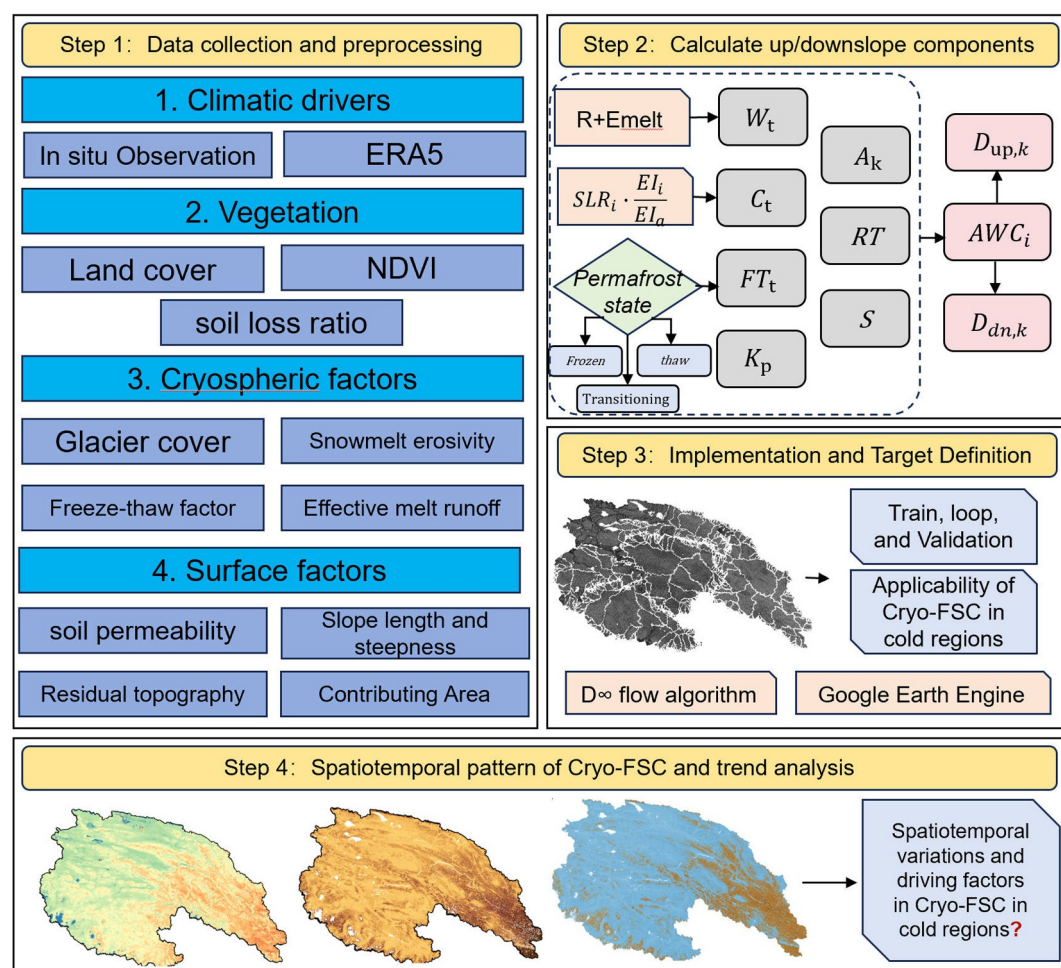


Figure 2. Flowchart of the Cryo-FSC computation framework. Each index abbreviation and the resulting vector output files are described in detail within this study. All input layers and the outputs from each computational module were normalized prior to integration.

To further support sediment and erosion analysis, we integrated supplementary data sets including soil (Liu et al., 2020, 2022), land cover (Xiao Zhang et al., 2021), vegetation changes (Wang et al., 2022), as well as land surface temperature (Xiaodong Zhang et al., 2021; Zhao et al., 2021), precipitation (Wang et al., 2018), runoff (X. Li et al., 2023; L. Wang et al., 2021), permafrost (Qiu, 2012; Wang et al., 2009; Zhao et al., 2020), glacier (S. Wang et al., 2021; Yao et al., 2012) and snow distributions (Yan et al., 2022). The IC data were obtained from the Li et al. (2025a). Sediment data were sourced from a gridded sediment flux product for the SRYR developed by Li et al. (2025b). More details can be found in Figures S1–S3 and Table S1 in Supporting Information S1.

It is important to note that the spatial and temporal resolutions of these data sets were not fully consistent with the satellite-derived sediment data, and uncertainties existed between gridded products and field observations. To address these inconsistencies, we employed the Variance-Upscaled Quantile Mapping (VUQM) method developed by Ma et al. (2025), which enables the construction of spatially coherent gridded climate fields (e.g., precipitation and air temperature) for sub-basin scale correlation analysis. Sub-basin averages were then derived accordingly. Finally, all processed gridded data sets were incorporated into hierarchical spatial calculations of Cryo-FSC. During this process, Global Moran's I was applied to quantify spatial autocorrelation in both feature locations and attribute values, revealing clustered, dispersed, or random spatial patterns depending on the underlying spatial structure of sediment connectivity.

2.4. Trend Analysis and Attribution of Cryo-FSC Dynamics

To assess the temporal evolution of sediment connectivity, we applied the non-parametric Mann–Kendall trend test and Pearson correlation analysis to explore changes in Cryo-FSC over the SRYR. We examined the spatiotemporal coupling between Cryo-FSC and key environmental indicators across varying cryospheric factors, vegetation conditions, and ground thermal states (e.g., land surface temperature in permafrost and seasonally frozen ground zones). This enabled us to explore how landscape and cryosphere heterogeneity modulate sediment connectivity patterns. To identify and quantify the drivers of Cryo-FSC variability, we employed Partial Least Squares Structural Equation Modeling (PLS-SEM) using the SmartPLS software platform.

It is worth noting that to mitigate the inflation of statistical significance caused by spatial autocorrelation, the correlation and PLS-SEM analyses were conducted at the sub-basin scale rather than the fine-pixel scale. This spatial aggregation effectively alleviates spatial dependence (Z. Li et al., 2025). Furthermore, as a non-parametric variance-based approach, PLS-SEM is relatively robust to residual multi-collinearity and non-independent observations commonly found in georeferenced environmental data sets. The latent variables incorporated into the model included: climatic drivers, such as precipitation (P) and air temperature (T); cryospheric factors, including the freeze-thaw index (FT) and the contribution of snow/ice melt to water-induced erosion (SM); surface factors, such as land use and geomorphometric metrics; vegetation factors.

Manifest variables used to represent these constructs included mean air temperature (T), precipitation (P), vegetation cover (V), land surface temperature (LST), runoff (Q), Glacier cover (G), freeze-thaw index (FT), and snowmelt-derived discharge (SM). Geomorphological indicators included slope (S), drainage area (A), normalized river channel steepness (K_{sn}), peak ground acceleration (PGA), and lithology erodibility (L). Each latent variable was characterized by one or more observed indicators. To assess model performance, we evaluated the Standardized Root Mean Square Residual (SRMR) and Normed Fit Index (NFI), as provided by SmartPLS. A good model fit was indicated by SRMR < 0.08 and NFI > 0.90, following commonly accepted criteria.

3. Results

3.1. Applicability and Dynamic Characters of the Cryo-FSC Index

The Cryo-FSC index reveals a portrait of sediment connectivity with far greater temporal detail and a wider spatial dynamic range across the SRYR compared to the static IC index (Figure 3). While the IC provides a generalized map of potential sediment transport pathways (Figure 3a), the dynamic Cryo-FSC exhibits significantly greater spatio-temporal heterogeneity, pinpointing transient hotspots of high connectivity that are directly coupled to cryospheric and climatic drivers (Figure 3b). This heightened detail is not merely resolution in time and seasons but reflects the explicit integration of sediment transport processes. As illustrated in the cross-sectional comparison in the middle sections of SRYR (Figures 3c and 3d), Cryo-FSC operates over a wider dynamic value range in sediment connectivity, more effectively distinguishing between transport-efficient river valleys and transport-limited hilltops, a distinction that remains muted in the static IC.

The functional validity of Cryo-FSC is demonstrated by its strong intra-annual coupling with key hydrological and thermal drivers (Figure 4). To test the temporal performance of Cryo-FSC and its response to climatic and cryospheric drivers, we analyze a complete hydrological year—spanning one full active-layer melt-freeze cycle from May 2015 to April 2016. We find that the basin-weighted Cryo-FSC time series exhibits distinct seasonal pulses, with peak connectivity occurring during the primary thaw season (June to September). This peak aligns with periods of maximum land surface temperature (LST), precipitation, and dynamic snowmelt (Figures 4a–4c). Crucially, the model reveals a critical process linkage, wherein the peak in Cryo-FSC consistently precedes the observed peak in suspended sediment flux (SSF) at the basin outlet by approximately 3–5 days (Figure 4c). This temporal lead suggests that Cryo-FSC successfully captures the expansion and subsequent mobilization phases of sediment transport pathways, endowing the index with predictive power for identifying periods of high sediment delivery potential. In addition, we found a distinct thermal threshold that governs the activation of basin-wide sediment connectivity (Figure 4d). We find that functional connectivity remains low and largely decoupled when LST is below a critical threshold of $\sim -1.1^{\circ}\text{C}$. Above this threshold, however, Cryo-FSC becomes strongly and positively correlated with LST ($R^2 = 0.37$, $P < 0.05$), signifying a regime shift from a transport-limited, frozen state to a thermally-activated, transport-efficient thawed state. It indicates that Cryo-FSC can capture a fundamental property of cryospheric fluvial systems—a sharp, thermally-driven transition from a disconnected to

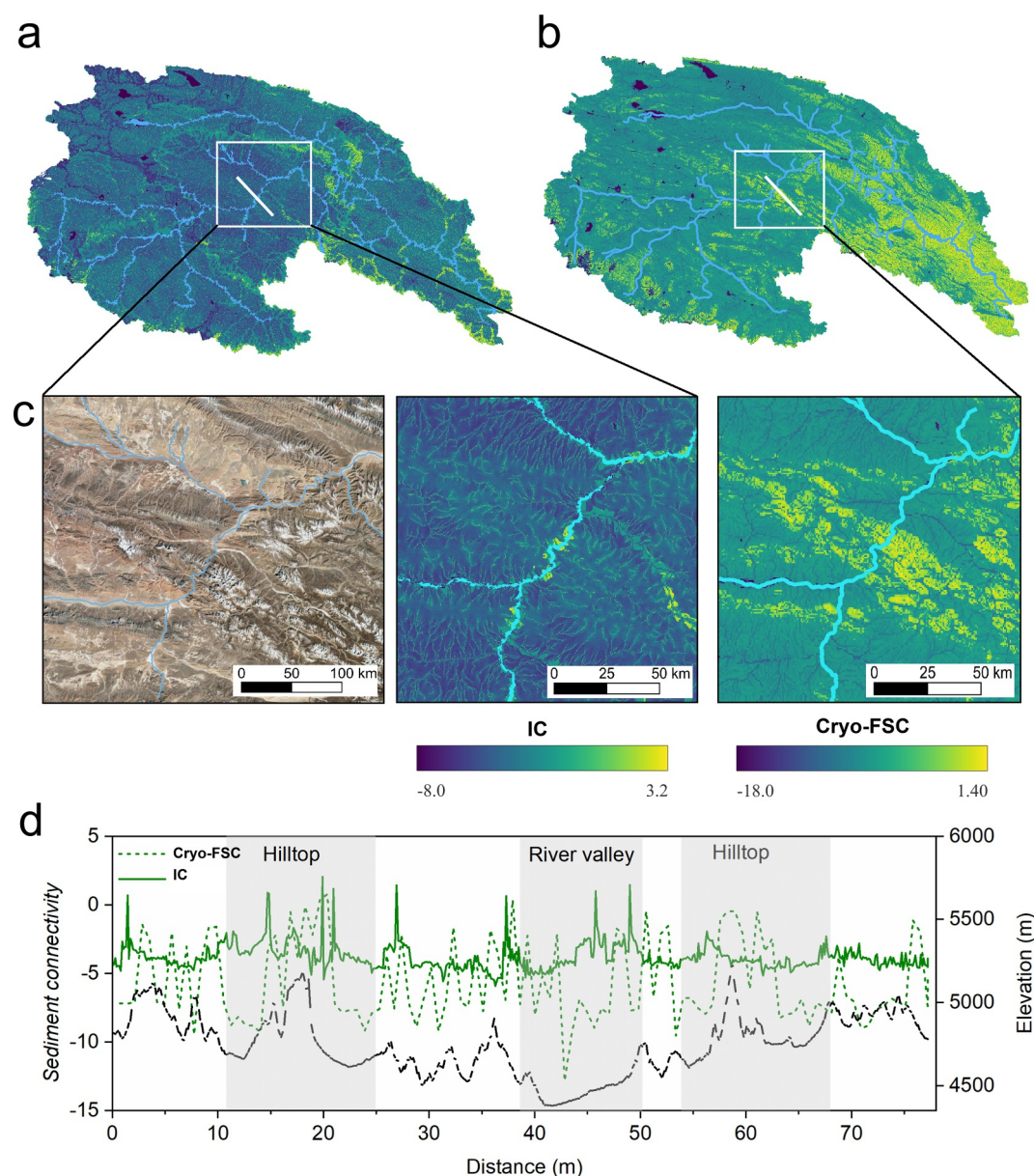


Figure 3. Comparison of structural (IC) and functional (Cryo-FSC) connectivity in the Source Region of the Yangtze River (SRYR). (a) The static IC provides a baseline, time-invariant representation of spatial connectivity, recalculated here from a 10m DEM based on the framework of Li et al. (2025a). (b) Visualization of the dynamic Cryo-FSC (30 m resolution), representing the annual mean connectivity from 2000 to 2022. Cryo-FSC spatially accentuates the heterogeneity of hydro-sediment connectivity and captures its spatio-temporal dynamics. (c) A magnified view of a tributary confluence with the Tongtian River, where a side-by-side comparison with satellite imagery demonstrates that Cryo-FSC more accurately resolves the fine-scale connectivity of the fluvial network and adjacent hillslopes. (d) Comparison of elevation, IC, and Cryo-FSC along a cross-section of the SRYR mainstream. Although the higher-resolution DEM for IC may provide more spatial detail, Cryo-FSC exhibits a significantly wider value range in its distribution, better representing the spatio-temporal changes of regional hydro-sediment connectivity.

a connected state, dictated by the changes in cryospheric factors. Conversely, Cryo-FSC exhibits a significant negative correlation with snow depth ($R^2 = 0.44$, $P < 0.05$), correctly representing the snow cover and frozen ground can suppress surface erosion and dampen connectivity in the frozen season (Figure 4e). Collectively, these results validate Cryo-FSC as a robust tool that not only maps potential pathways but also quantifies their dynamic behavior in response to the unique drivers of permafrost-dominated river systems.

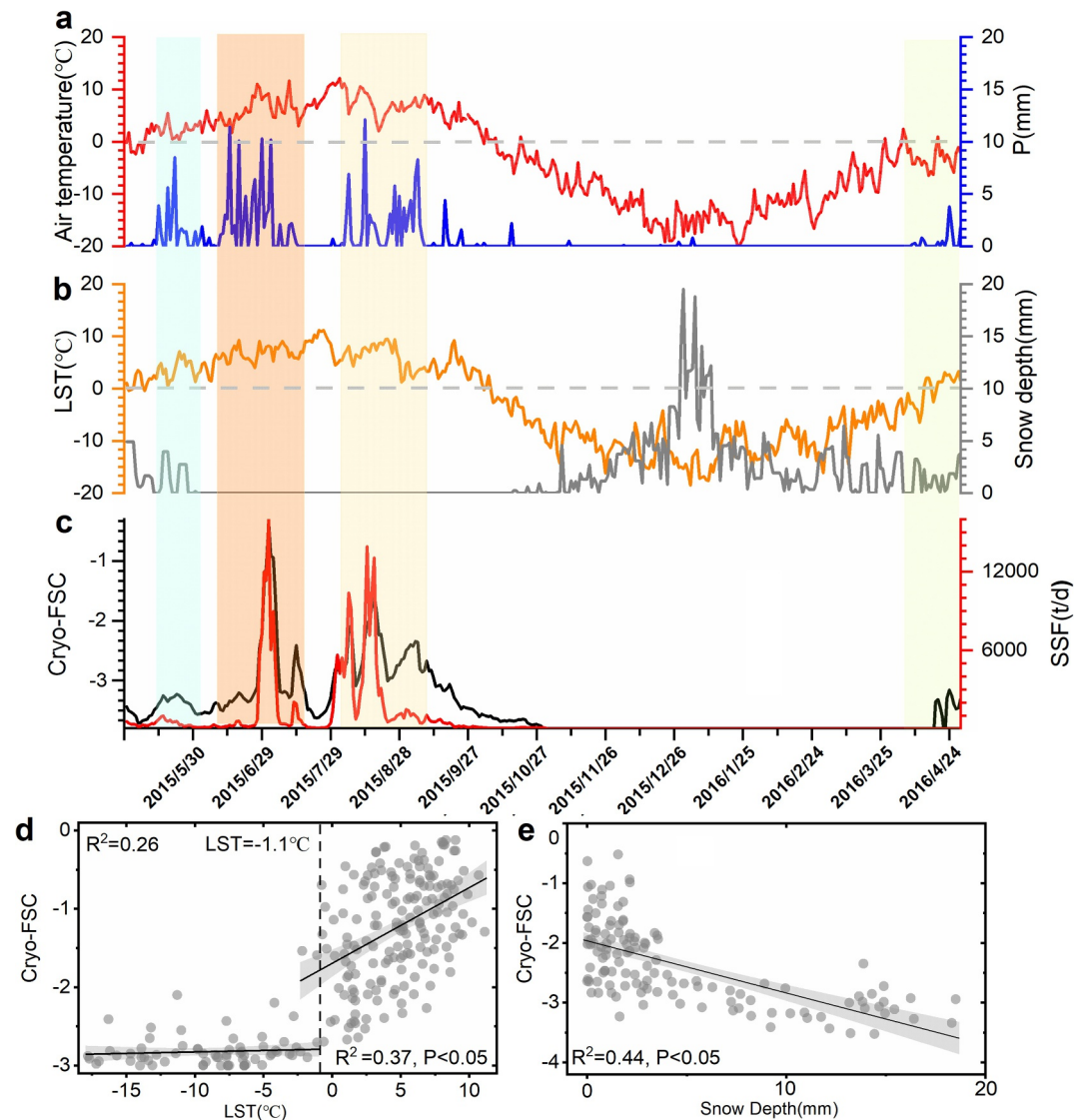


Figure 4. Intra-annual dynamics of Cryo-FSC and its coupling with climatic, cryospheric factors, and sediment flux. The analysis examines a complete hydrological year from May 2015 to April 2016. (a) Annual dynamics of air temperature and precipitation (P). (b) Annual dynamics of Land Surface Temperature (LST) and snow depth. (c) Dynamics of riverine suspended sediment flux (SSF) and basin-weighted Cryo-FSC, showing an approximately 3–5 days lag where SSF follows the Cryo-FSC signal. (d) The threshold effect between LST and Cryo-FSC. When $LST > -1.1^{\circ}\text{C}$, the basin's hydro-sediment connectivity is activated, leading to a significant positive correlation with LST ($P < 0.05$). (e) The negative correlation between snow depth and Cryo-FSC, where a frozen, snow-covered surface implies a cessation of hillslope erosion and sediment production dynamics. The three shaded bands on the left highlight the moments when both SSF and Cryo-FSC reach their peaks. The shaded band on the far right signifies a period during snowmelt when surface processes and hydro-sediment connectivity are activated, yet this is not immediately reflected in the SSF.

Beyond demonstrating intra-annual mechanism linkages, we explicitly validated the predictive capacity of the Cryo-FSC index using long-term regression analysis against observed hydro-sedimentological variables at the basin outlet spanning 2000–2022 (Figure 5). At the monthly scale, Cryo-FSC demonstrates a highly significant exponential relationship with both observed sediment fluxes ($R^2 = 0.58$, $P < 0.001$) and average runoff discharge ($R^2 = 0.74$, $P < 0.001$) (Figures 5a and 5b). This confirms that the index effectively captures the extreme intra-annual seasonality characteristic of cold-region sediment transport—a temporal dynamic that static topographical indices are fundamentally incapable of reproducing. Furthermore, at the annual scale, the annual mean Cryo-FSC significantly explains the variance in total suspended sediment fluxes ($R^2 = 0.38$, $P = 0.01$), drastically outperforming the traditional static IC ($R^2 = 0.07$, $P = 0.52$) (Figures 5c and 5e). A similarly stark performance gap is

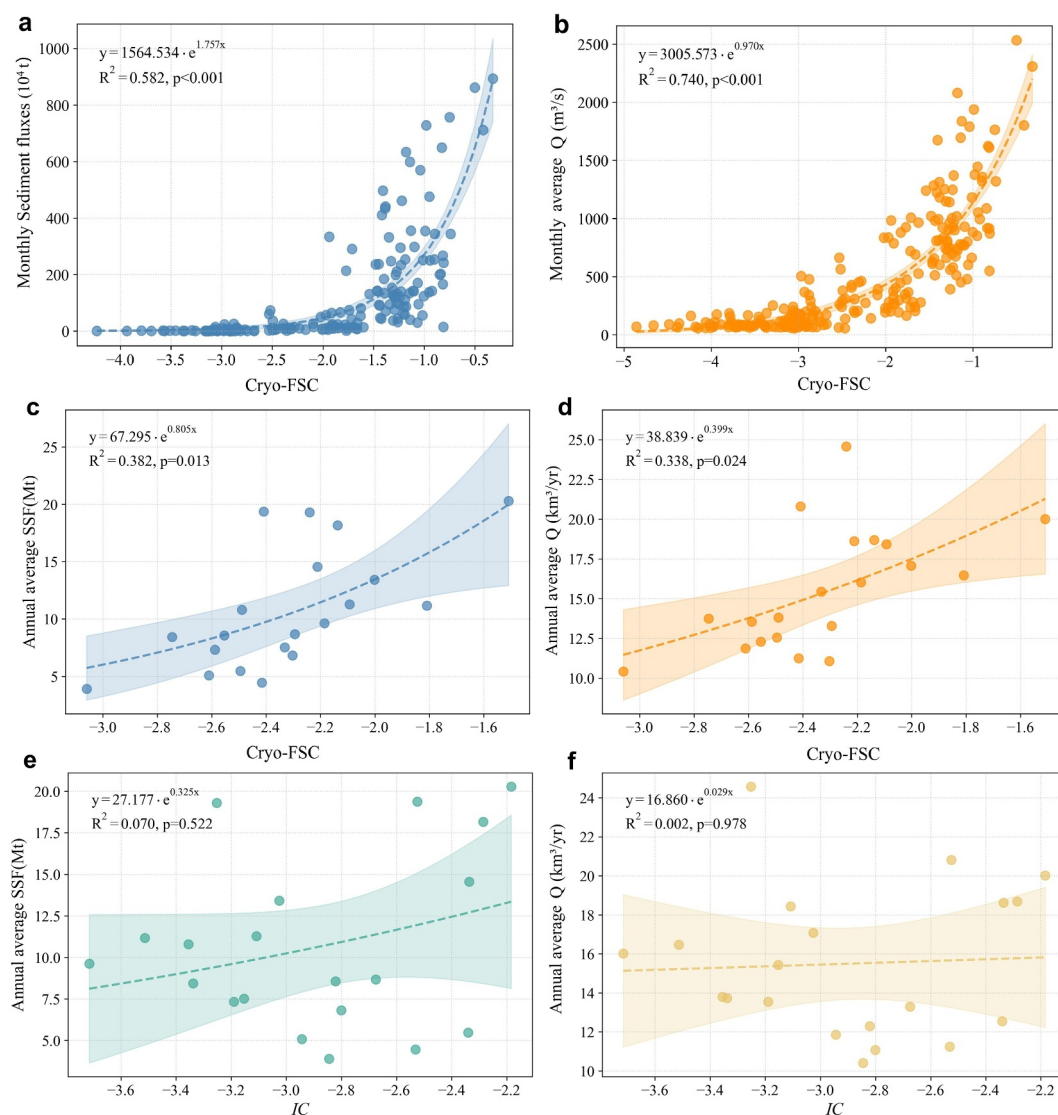


Figure 5. Non-linear regression relationships between observed hydro-sedimentological variables and connectivity indices at the basin outlet (2000–2022). (a) Monthly sediment fluxes (10^4 t) versus the dynamic Cryo-FSC index. (b) Monthly average water discharge (Q) versus the Cryo-FSC index. (c) Annual average suspended sediment flux (SSF, Mt) versus the annual mean Cryo-FSC index. (d) Annual average Q (km^3/yr) versus the annual mean Cryo-FSC index. (e–f) The relationships between the static structural connectivity index (IC) and annual average SSF and Q, respectively. Note that the static IC values utilized here are derived from a previously developed product based on the High Mountain Asia DEM (Li et al., 2025a), which covers only a partial area of the Source Region of the Yangtze River (SRYR). The dashed lines represent the best-fit exponential regression models, with equations, coefficients of determination (R^2), and P -values provided for each panel. The shaded regions indicate the 95% confidence intervals of the regression fits.

observed for predicting annual discharge (Figures 5d and 5f). These comparative regression results directly answer the critical need for dynamic connectivity modeling, proving that integrating cryospheric and climatic transients into the Cryo-FSC framework translates directly into a superior quantitative indicator of actual sediment delivery.

3.2. Spatial Heterogeneity of Cryo-FSC

The spatial distribution of the annual mean Cryo-FSC across the SRYR reveals a pronounced east-west gradient, which contrasts sharply with the patterns suggested by static IC (Figures 3 and 6). A dominant feature is the markedly higher connectivity in the eastern portion of the basin (Tongtian River section and Zhimenda),

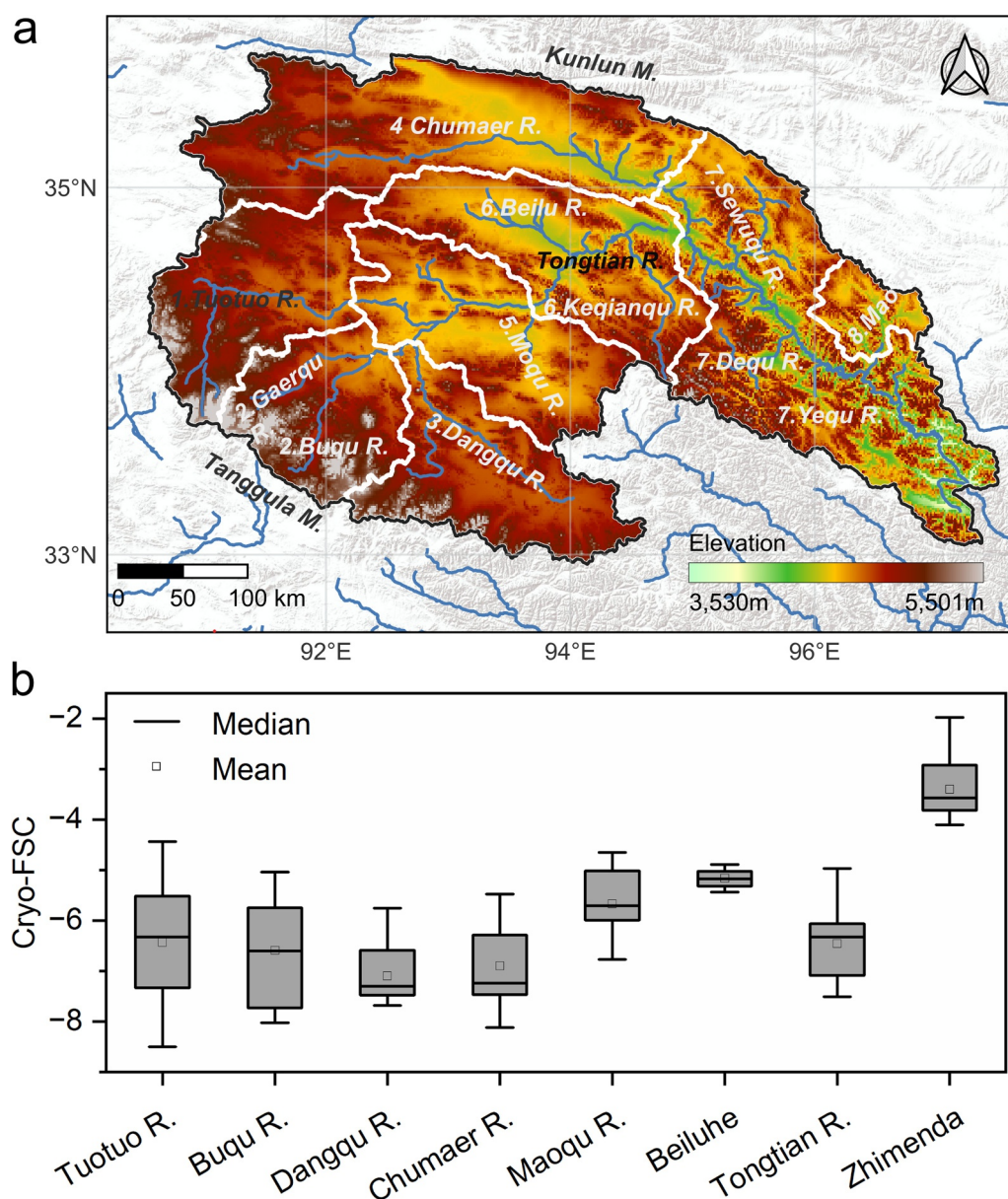


Figure 6. Spatial variability of basin-weighted Cryo-FSC in the Source Region of the Yangtze River (SRYR). (a) Digital Elevation Model (DEM) of the SRYR, illustrating the eight major sub-basins delineated based on hydrological processes to facilitate statistical analysis. (b) Boxplot distribution of Cryo-FSC values for the eight major sub-basins. A clear sequential increase in Cryo-FSC is evident from the headwaters in the west (source) to the outlet in the east (sink).

compared to the relatively lower connectivity in the western headwaters (Figure 6b). High Cryo-FSC values in the east are concentrated along the river valleys of the mainstream and its major tributaries, such as the Beiluhe and Chumaer Rivers. This heightened connectivity is closely associated with the region's steeper topographic gradients and the prevalence of seasonally frozen ground (Figure 7), which is inherently more susceptible to erosion compared to the contiguous permafrost that dominates the western plateau (Figure 6a).

We also identified distinct hotspots of high Cryo-FSC that are strongly coupled with zones of peak sediment transport. These hotspots are primarily located in two key geomorphic settings: (a) the western proglacial zones, where meltwater channels are spatially associated with proglacial routing corridors, facilitating efficient sediment evacuation from the Tanggula Mountains (e.g., in the Gaerqu and Buqu River headwaters); and (b) the major confluence zones in the downstream part of the basin, which act as sinks for the entire watershed's runoff

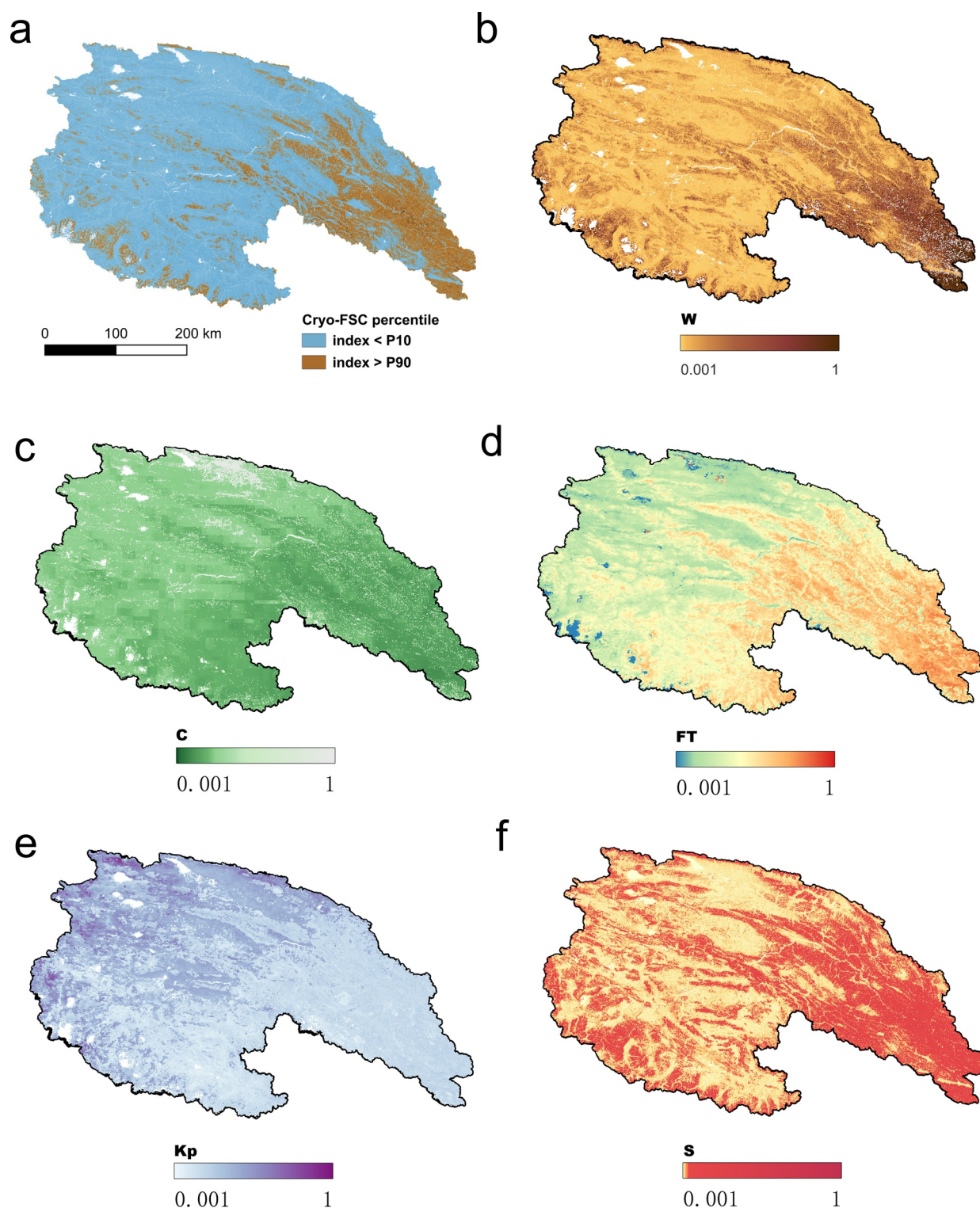


Figure 7. Spatial heterogeneity of the Cryo-FSC index and its constituent factors. Maps at a 30-m resolution show the normalized, multi-year mean (2000–2022) of: (a) Cryo-FSC percentiles, highlighting the lowest (<P10) and highest (>P90) connectivity zones; (b) Water Erosivity Factor (W); (c) Dynamic Vegetation Factor (C); (d) Freeze-Thaw Factor (FT); (e) Intrinsic Soil Permeability Factor (K_p); and (f) Slope Gradient (S).

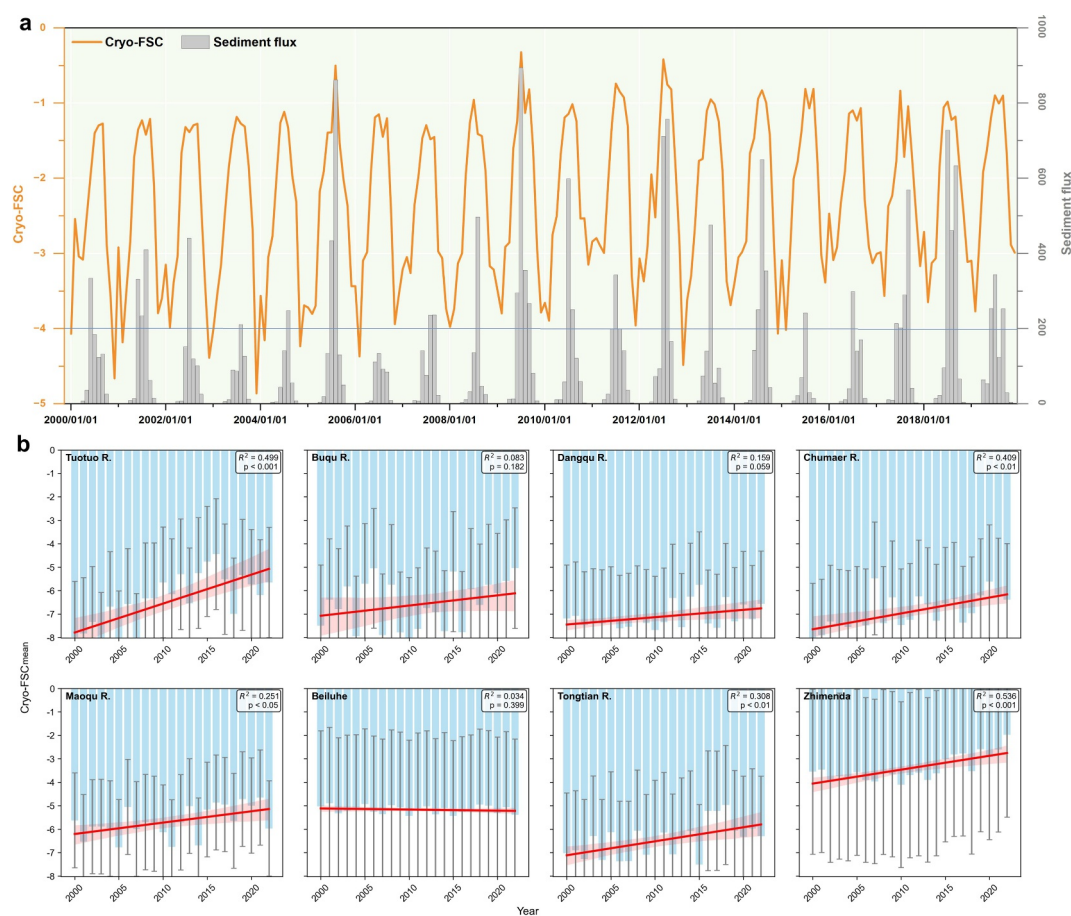


Figure 8. Inter- and intra-annual dynamics of Cryo-FSC. (a) Time-series variations of basin-weighted Cryo-FSC and the observed sediment flux (10^4 t) changes from 2000 to 2020, which are sourced from the Zhimenda hydrological station. (b) Annual mean Cryo-FSC variations and trends across the eight major sub-basins. The downward-extending light-blue bars represent the annual mean Cryo-FSC values for each sub-basin, with the gray error bars indicating the standard deviation to reflect intra-annual variability. The solid red lines denote the linear trends of the annual means, and the red shaded areas represent the 95% confidence intervals of these linear regressions.

(Figure 6). In contrast, across the vast permafrost-dominated landscapes of the upper and middle reaches, high Cryo-FSC values are conspicuously confined to the near-channel and riparian zones (Figure 3c). The intense connectivity along these narrow corridors implies that sediment from these zones can be efficiently routed into the fluvial system, a process whose contribution may have been underestimated in studies relying solely on basin-outlet or hillslope-focused measurements (Kanevskiy et al., 2016; J. Li et al., 2024).

3.3. Inter- and Intra-Annual Variability of Cryo-FSC

The Cryo-FSC index exhibits significant inter-annual variability and spatially divergent trends across the SRYR (Figure 8). In 2000–2022, the basin-weighted annual mean Cryo-FSC for the entire SRYR shows a general increasing trend. It is important to note that this modeled intensification of functional connectivity aligns strongly with the increase in riverine sediment flux in the region, as evidenced by both long-term in situ hydrological observations (Li et al., 2020) and satellite-derived sediment flux products (J. Li et al., 2025b) (Figure 8a and Figures S5 and S8 in Supporting Information S1). However, this overall trend masks substantial spatial heterogeneity among the eight major sub-basins. Notably, the glacierized sub-basins (where glacier coverage >2%) in the western headwaters, such as Tuotuo River (with a growth rate of $12.64 \pm 3.83\%/yr$, $P < 0.001$) and Buqu River ($4.35 \pm 1.52\%/yr$), display pronounced positive trends in Cryo-FSC, coinciding with increasing sediment flux. In contrast, sub-basins like the Dangqu River ($3.13 \pm 0.94\%/yr$) show non-significant increasing trends, suggesting that its well-developed vegetation effectively impedes the expansion of sediment pathways

(Figure 7c). For areas characterized by steeper topographic gradients and significant permafrost (seasonal frozen ground) degradation (Figure S4 in Supporting Information S1), such as the Chumaer River ($6.77 \pm 1.41\%/yr$, $P < 0.01$) and Zhimenda ($5.93 \pm 2.37\%/yr$, $P < 0.001$), statistically significant increases in Cryo-FSC are also observed. This spatial divergence underscores the complex, non-uniform response of sediment connectivity to regional climatic and cryospheric changes, with areas undergoing more rapid permafrost degradation and glacial retreat likely experiencing accelerated increases in sediment transport. The general upward trend in Cryo-FSC for the SRYR suggests an increasing efficiency of sediment mobilization and transport over the past two decades, potentially leading to the enhanced sediment flux observed at the Zhimenda outlet and downstream river networks (Figure S5 in Supporting Information S1; Zhang et al., 2020).

On an intra-annual scale, Cryo-FSC in the SRYR demonstrates distinct seasonal pulses, primarily dictated by the thermal regime and hydrological forcing (Figures 4 and 8a). We also observed that the basin-weighted Cryo-FSC peaks sharply during the thaw period, align with the peak sediment flux (Figure 4c), which is consistent with the maximum LST and increased precipitation (Figures 4a, 4b, and 8a). During the freezing period, surface erosion processes are largely halted due to frozen ground, leading to a cessation of riverine sediment transport (Lamoureux et al., 2014). However, we found that despite Cryo-FSC being at its annual lowest during this time, it can rapidly respond with an upward trend once surface temperatures rise. Erosion is initiated (mostly during spring snowmelt erosion) (Figure 8a). The three shaded bands in Figure 4c highlight the moments of peak SSF and Cryo-FSC. In contrast, the far-right shaded band marks a period during snowmelt when initial surface processes and connectivity are activated, though not yet fully reflected in riverine SSF measurements, indicating the pre-conditioning role of Cryo-FSC.

3.4. Driving Factors of Cryo-FSC

We found that cryospheric elements and surface factors exert paramount control over the dynamics of sediment connectivity in permafrost-dominated river systems (Figures 8a and 8b). The PLS-SEM model highlights that cryospheric factors (G, FT, SM) and surface factors (LST, Q, S) exhibit the strongest direct causal effects on Cryo-FSC during both thaw and frozen periods. Specifically, in the thaw period, cryospheric factors (with a path coefficient of 0.33, $P < 0.05$) and surface factors (with a path coefficient of 0.56, $P < 0.05$) are significant positive drivers. This is further substantiated by the Pearson correlation analysis (r , Figure 9b), where glacier cover shows the highest positive correlation with Cryo-FSC ($r = 0.85$, $P < 0.01$), signifying that glacial meltwater erosion provides an abundant water supply and erosion potential. Similarly, slope ($r = 0.55$, $P < 0.01$) and snowmelt-derived discharge (SM) ($r = 0.52$, $P < 0.01$) are strong positive correlates, reflecting their direct roles in generating runoff and mobilizing sediment across steep terrains. The pervasive influence of these factors underscores a fundamental difference from temperate regions. Sediment connectivity in cold environments is intricately tied to thermal regimes and the unique erosional processes associated with permafrost degradation and glacial activity, which largely dictate the timing and magnitude of sediment mobilization and transport efficiency (D. Li et al., 2024; Zhang et al., 2023).

Beyond the direct drivers, climate and vegetation interact in complex feedback loops to modulate sediment pathways, with distinct mechanisms operating during thaw and frozen periods (Figure 9a). During the thaw period, climatic drivers (T, P) exert an indirect influence on Cryo-FSC, primarily by enhancing surface factors (e.g., increased LST, Q, SM), which then propagate their effects to connectivity. Furthermore, our model shows a significant negative path coefficient from vegetation to Cryo-FSC (with a path coefficient of -0.26), indicating that a denser vegetation cover effectively dampens sediment connectivity during this season. Conversely, during the frozen period, the direct statistical influence of vegetation on Cryo-FSC diminishes substantially (with a path coefficient of -0.02). However, cryospheric factors (with a path coefficient of 0.52) and surface factors (with a path coefficient of 0.81, $P < 0.01$) maintain strong control over the connectivity index. This phase-dependent interaction underscores how climate-driven shifts in vegetation cover, coupled with the seasonal transition of active layer conditions, dynamically restructure the efficiency and pathways of sediment mobilization and transport in cold regions.

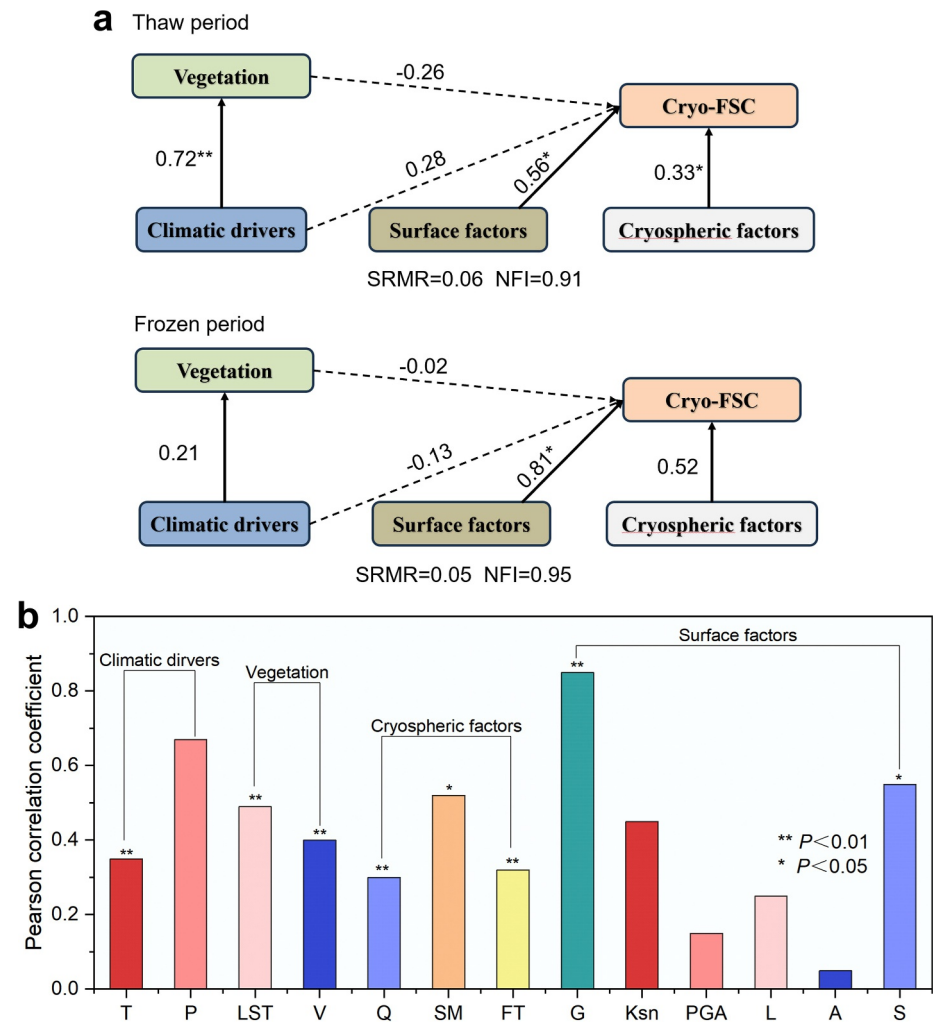


Figure 9. Attribution and statistical analysis of the driving factors of Cryo-FSC. (a) Partial Least Squares Structural Equation Model (PLS-SEM) illustrates the causal effects among vegetation, climatic drivers, surface factors, cryospheric factors, and Cryo-FSC during the thaw (above) and frozen (below) periods from 2000 to 2022. Path coefficients are marked above the arrows. Significance levels are denoted by * ($P < 0.05$) and ** ($P < 0.01$). Solid lines indicate significant path coefficients, while dotted lines represent insignificant ones. (b) Statistical analysis detailing the control of environmental variables on Cryo-FSC. Variables are categorized into four groups, including mean air temperature (T), precipitation (P), vegetation cover (V), land surface temperature (LST), runoff (Q), glacier cover (G), freeze-thaw index (FT), and snowmelt-derived discharge (SM). Geomorphological indicators comprise slope (S), drainage area (A), normalized river channel steepness (Ksn), peak ground acceleration (PGA), and lithology erodibility (L).

4. Discussion

4.1. Cryo-FSC Better Reflects the Relationship in Sediment Source-to-Sink Processes in Cold Regions

The Cryo-FSC index provides a robust spatial framework for linking hillslope erosion processes to their potential transport pathways—a relationship that is particularly complex in cryosphere-dominated landscapes. On a macro-scale, the annual mean Cryo-FSC distribution (2000–2022) exhibits a strong spatial coupling with observed patterns of riverine suspended sediment concentration (SSC) and discharge across the SRYR (Figure 10a). Furthermore, while the Cryo-FSC model does not explicitly simulate the physical initiation of localized erosion features such as retrogressive thaw slumps (RTS), overlaying observed RTS inventories onto the connectivity map allows us to reasonably infer their downstream impacts. Specifically, it highlights that RTS located within high-connectivity zones are primed to contribute substantially more sediment to the fluvial system compared to disconnected counterparts (Kokelj et al., 2013; Luo et al., 2022; Ma et al., 2025).

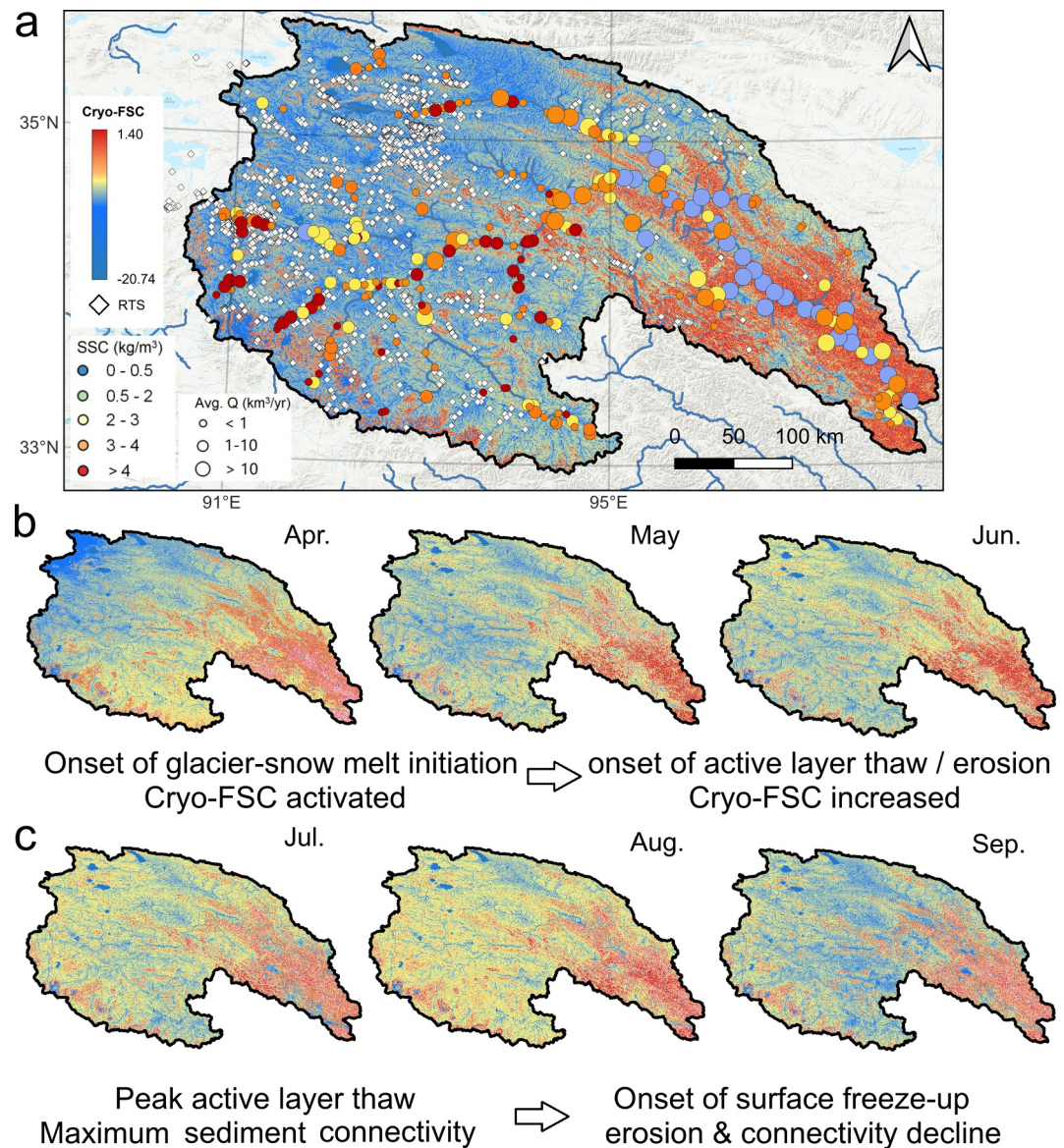


Figure 10. Linking Cryo-FSC with sediment dynamics and source-to-sink processes in the source region of the Yangtze River (SRYR). (a) Annual mean Cryo-FSC map (2000–2022) for the SRYR, showing its spatial coupling with riverine suspended sediment concentration (dot sizes indicate annual mean discharge). (b, c) Monthly mean Cryo-FSC visualizations from April to September (2000–2022), illustrating the four distinct phases of erosion and connectivity within the year. For more detailed visualizations of specific sediment dynamics, please refer to Figures S5 and S6 in Supporting Information S1 and Li et al. (2023, 2025b).

Spatially, our analysis shows that RTS are predominantly concentrated in the western headwaters in the SRYR, regions characterized by extensive permafrost and lower Cryo-FSC values. It also indicated a substantial number of retrogressive thaw slumps are not effectively connected to the river channels or active erosion gullies (e.g., in the uppermost reaches of the Chumaer River and the first bend of the Tuotuo River). This implies that these dislodged sediments are prone to being transported by torrential rains during the flood season, only to become trapped and deposited at the footslopes or within small gullies, failing to reach the downstream areas (J. Li et al., 2024). However, sediment generated from RTS in highly connected locations (particularly in proglacial zones, along incised riverbanks, or the downstream valleys) can be efficiently transferred directly into the fluvial system, leading to locally significant increases in sediment flux that are markedly higher than in those poor-connected regions. This finding suggests that Cryo-FSC can effectively contextualize the impact of localized,

high-magnitude erosion events like RTS, explaining why some slumps contribute directly to basin-scale sediment budgets while others may remain locally confined.

Beyond its spatial explanatory power, Cryo-FSC excels at capturing the intra-annual evolution of erosion and sediment transport pathways, which are tightly governed by the seasonal progression of freeze-thaw cycles (Figures 10b and 10c). The index delineates four distinct phases within a hydrological year. The initial phase (b1→b2, in Figure 10) is characterized by the onset of glacier-snowpack melt, which initiates erosional activity by providing the first major pulse of surface runoff (Beylich et al., 2017). As the season progresses and snow cover diminishes (b2→b3, in Figure 10), the primary driver of connectivity shifts to the thawing of the active layer. During this thaw period, the dampening effect of vegetation on Cryo-FSC mechanistically represents the enhancement of mechanical soil cohesion via root systems and the protective interception of rainfall kinetic energy, which together stabilize slopes and reduce runoff erosivity (D. Li et al., 2024). Peak Cryo-FSC occurs during the period of maximum active layer thaw (c1→c2, in Figure 10), corresponding to the seasonal zenith of sediment connectivity and peak sediment availability from hillslope sources (Wang et al., 2009).

Finally, as the surface begins to freeze (c2→c3, in Figure 10), both erosion and connectivity sharply decline. However, even during these frozen or transitional periods, our framework captures sustained or episodic connectivity that we attribute to localized mechanisms not explicitly modeled but inferred from regional observations. Specifically, transient warming events may activate sediment delivery through snowmelt pulses, intensive scour from river ice during ice jams and breakup events, and the activation of shallow sub-surface flow or accelerated bank erosion (Beltaos & Burrell, 2021; J. Li et al., 2023). This detailed seasonality of connectivity, which links specific cryospheric processes to the functional state of the sediment transport system, demonstrates that Cryo-FSC is not merely correlated with sediment flux but mechanistically reflects the seasonal waxing and waning of the relationship between hillslope sediment production and its potential for in-channel expression.

4.2. Limitations and Future Perspectives of Cryo-FSC

While the Cryo-FSC index represents a significant conceptual advance for quantifying sediment dynamics in cold regions, its application is subject to inherent limitations dictated by scale and process representation. First, the index's fidelity is constrained by the spatial resolution of its input data (e.g., 30 m DEM and satellite-derived products). Critical cryospheric processes, such as the initiation of thermal erosion gullies, retrogressive thaw slumps, and the dynamics of subglacial conduits, often occur at sub-grid scales (Lewkowicz & Way, 2019). Consequently, while Cryo-FSC effectively models connectivity across the established landscape, it may not fully capture the genesis of new, fine-scale pathways that can abruptly alter sediment routing. Second, our current framework is fundamentally two-dimensional, focusing on surface and near-surface lateral connectivity. It does not explicitly account for vertical connectivity, such as the crucial interactions between surface water, supra-permafrost groundwater, and the development of taliks beneath thermokarst lakes (D. Li et al., 2024; Ma et al., 2025; Walvoord and Kurylyk, 2016). These vertical fluxes can significantly modify subsurface thermal regimes and hydrological pathways, thereby influencing surface sediment availability in ways not yet captured by the index. Future research should therefore aim to integrate multi-scale modeling approaches or develop parameterization schemes to incorporate the effects of these sub-grid and vertical processes.

A secondary challenge involves the integration of multi-source data sets with diverse resolutions and accuracies. To systematically mitigate input uncertainties, we implemented a rigorous quality-control framework. Climatic forcing biases were minimized using VUQM and Triple Collocation methods (Ma et al., 2025). Spatial mismatches between physical simulations and geomorphic routing were resolved via strict downscaling protocols. Furthermore, rather than relying on disparate remote sensing products, core cryospheric drivers were constrained by the DHTC model and cross-validated against satellite ensembles (Figures S7–S9 in Supporting Information S1). Although residual sub-grid uncertainties inevitably persist, this integration framework ensures the Cryo-FSC trends remain statistically robust.

The final frontier for improvement lies in refining the model's internal structure and its integration into broader predictive frameworks. The Cryo-FSC, while physics-informed, relies on several parameterized factors (e.g., the temperature-index for meltwater generation, empirical thresholds for the freeze-thaw factor) which may not fully capture the non-linear responses inherent in cryospheric systems (Deng et al., 2025; McVey et al., 2023). A promising direction would be to couple the Cryo-FSC framework with fully distributed, physics-based energy balance and soil thermal models, allowing for a more mechanistic representation of meltwater generation and

active layer dynamics (Grenier et al., 2018). Furthermore, this study, by design, focuses on functional connectivity driven by short-term climatic and hydrological variability. A truly holistic understanding of landscape evolution, however, requires bridging the gap between these transient functional dynamics and the long-term structural evolution of the landscape, which is governed by tectonic and geomorphic legacies (Heckmann et al., 2018). The ultimate goal is to evolve Cryo-FSC from a powerful diagnostic tool that explains current and past dynamics into a fully prognostic component within coupled landscape evolution models (Tucker and Slingerland, 1997), capable of projecting the trajectory of sediment cascades in these globally significant, rapidly transforming cold regions under future climate scenarios.

5. Conclusions

This study introduces the Cryo-FSC index, a novel tool designed to dynamically quantify functional sediment connectivity in permafrost-dominated river systems, directly addressing the limitations of static indices in cold regions. By explicitly incorporating key cryospheric and surface processes, Cryo-FSC provides unprecedented insights into the spatio-temporal evolution of sediment transport in the Source Region of the Yangtze River (SRYR). Our findings reveal a pervasive increasing trend in Cryo-FSC over the past two decades, particularly in glacierized headwaters, reflecting the intensified erosion driven by regional warming and permafrost degradation. The index adeptly captures the distinct intra-annual seasonality of sediment connectivity, characterized by a sharp activation above a critical land surface temperature threshold and a consistent lead time in predicting downstream sediment pulses. Furthermore, Cryo-FSC enhances our understanding of the differential impact of localized erosion events, such as retrogressive thaw slumps, by distinguishing between connected sediment sources that directly contribute to riverine flux and those that remain hydrologically isolated. This innovative framework offers a robust and adaptable methodology for assessing and forecasting sediment dynamics in rapidly changing cryospheric landscapes globally. Future work will focus on integrating Cryo-FSC with fully physics-based models and multi-scale observation systems to further refine its predictive capabilities and contribute to holistic landscape evolution modeling.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

All original data and codes used in this manuscript are available at the Zenodo data portal (<https://zenodo.org/records/19477605>) and are cited and referenced in the manuscript. The IC data set was developed by J. Li et al. (2025a). Boundaries of glaciers and permafrost are obtained from Zhao et al. (2021). The fifth-generation ECMWF Reanalysis (ERA5) data set is downloaded from Copernicus Climate Change Service (2019). The other used data can be found in Section 2.3 in the main text.

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